

## **Christopher Alvarez Diaz**

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### **PROFESSIONAL SUMMARY**

Results-driven Robotics Engineering student with a strong foundation in digital logic, RTL design, and systems integration. Hands-on experience developing and verifying a custom 8-bit CPU in Verilog and architecting autonomous software stacks in C++ and Python. Seeking a Hardware Design Verification Internship at AMD to leverage expertise in FPGA design, behavioral simulation, and hardware-software integration to validate next-generation architectures.

### **EDUCATION**

University of California, Santa Cruz - B.S., Robotics Engineering Expected Graduation: Jun 2029

### **TECHNICAL SKILLS**

**Hardware & Verification:** Verilog, RTL Simulation, AMD Vivado, FPGA Design, Digital Logic, Computer Architecture

**Software & Systems:** C++, Python, Bash/Linux CLI, Git/GitHub, ROS 2, Nav2, Vision AI

**Engineering Tools:** SolidWorks, OnShape, Gazebo Simulation, MATLAB

### **ENGINEERING PROJECTS**

#### **Custom 8-Bit Single-Cycle CPU | Verilog, AMD Vivado, RTL Verification | Aug 2026**

- Architected and implemented a complete 8-bit single-cycle processor from the gate level up, featuring a custom 12-instruction ISA with R/I/J-type encodings.
- Engineered the full datapath—including ALU, register file, control unit decoder, and memory modules—optimizing for critical path efficiency.
- Developed a comprehensive testbench suite, achieving a 100% pass rate in behavioral simulations using AMD Vivado to thoroughly verify logic operations and branch execution.

#### **Autonomous ROS 2 Payload Drone | ROS 2, C++, Python, Gazebo | Mar 2026 – Present**

- Developed an autonomous drone navigation software stack in C++ and Python, leveraging Nav2 and Vision AI for dynamic obstacle avoidance.
- Validated flight control algorithms through extensive simulated test flights across a virtual city environment in Gazebo prior to hardware deployment.
- Designed and assembled the physical drone platform utilizing CAD tools and soldering within a strict \$150 budget, bridging the gap between software simulation and hardware execution.

#### **Formula Slug (EV Racing Team) - Battery Sub-Team | Hardware Design | Sep 2026 – Present**

- Researched and designed bus bar configurations and actuator systems for a high-performance 120V EV battery pack using SolidWorks and OnShape & McMaster Carr.
- Optimized the battery layout architecture, successfully reducing the total size of the battery tray by 60%.
- Assembled and integrated low/high-voltage components, achieving a 100% pass rate during vehicle electrical testing.

### **PROFESSIONAL EXPERIENCE & RESEARCH**

#### **Undergraduate Biology Researcher | Antelope Valley College | Oct 2024 – Jun 2025**

- Investigated micro-arthropod behaviors in controlled environments, becoming the first high school student in the region to complete original research in collaboration with the college.
- Presented findings at the Spring Research Symposium and to a class of 32 students; research is currently in development for academic publication.

#### **Administrative and Financial Intern | UC Santa Cruz | Jan 2026 – Present**

- Processed and tracked ~150 campus business forms (requisitions, vendor registrations) per month, maintaining 100% compliance with university financial policies.
- Executed detailed budget projections for multiple departments to accurately determine resource allocation and improve financial efficiency.